

JaamSim: Simulation Tool Simplifying Complex Systems

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The rate at which technology is progressing today, one has to be creative, futuristic and code efficiently on the go. What if you could use your Java scripting skills to simplify the behaviour of complex systems? How about bringing it all down to models and simulations of sequences of well-defined events? This is what JaamSim lets you do!

JaamSim, short for Java Animation Modelling and Simulation, is a discrete event-simulation software package that can run on Windows, Linux or OS X systems, and is supported by Apache2.0 free open source software licence.

What to expect from discrete system-simulation software

The idea behind simulating a system is clear-cut—to replicate its functionality, imbibing all your ideas into it. The system is a bunch of components put together, each depicting a certain functionality at a certain time or under given conditions. The component could be a simulation entity, an entity generator or a source as it is colloquially called, a channel or entity conveyor, or an entity sink where the entire data converges.

The stimulus flows from one section, through the entire model, and finally appears at the sink, creating the process flow. This is, of course, the basic requirement for the simplest of systems. As complexity adds on, you might need a queue, a branch, additional resources or even an entity container to hold multiple entities.

Smooth processing. In JaamSim,

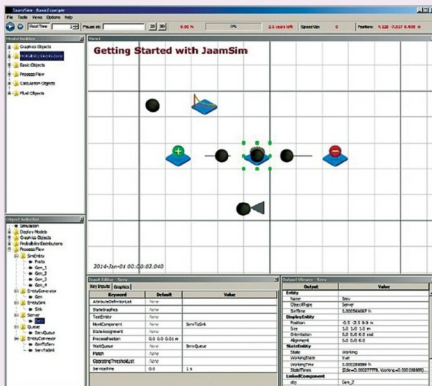


Fig. 1: Modelling a server

The checklist

- Java development environment like Eclipse
- System supporting OpenGL graphics version 2.1 and above
- JaamSim software from this month's DVD. Simply copy JaamSim executable file to your working environment and you are done

a sub-class of thread allows for entities to move through its methods, otherwise called a process. To create a process flow, you can move around, resize and place the components, connecting stages one after another, and also set the time at which each entity will be added to the model and the time each would spend at every stage of the same.

As each model is created,

Simulation Object stores inputs that define basic parameters such as run duration. While operating the system is as simple as selecting start/pause/stop/reset buttons from the control panel, the speed of simulation is decided by the real-time speed-up factor, as specified by you in the controls. By default, simulation time is equal to one real-time second. Multiple simulations can